

Prpgram 4 : integrate git th docker for source controlled application

Step 1 : make a directory and initialize npm and install express

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ npm init -y
Wrote to /home/1RV24MC093_SANDARSH_J_N/Downloads/devops/prg4/package.json:

{
  "name": "prg4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \\\"Error: no test specified\\\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ npm install express

added 68 packages, and audited 69 packages in 909ms

16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ ls
node_modules  package.json  package-lock.json
```

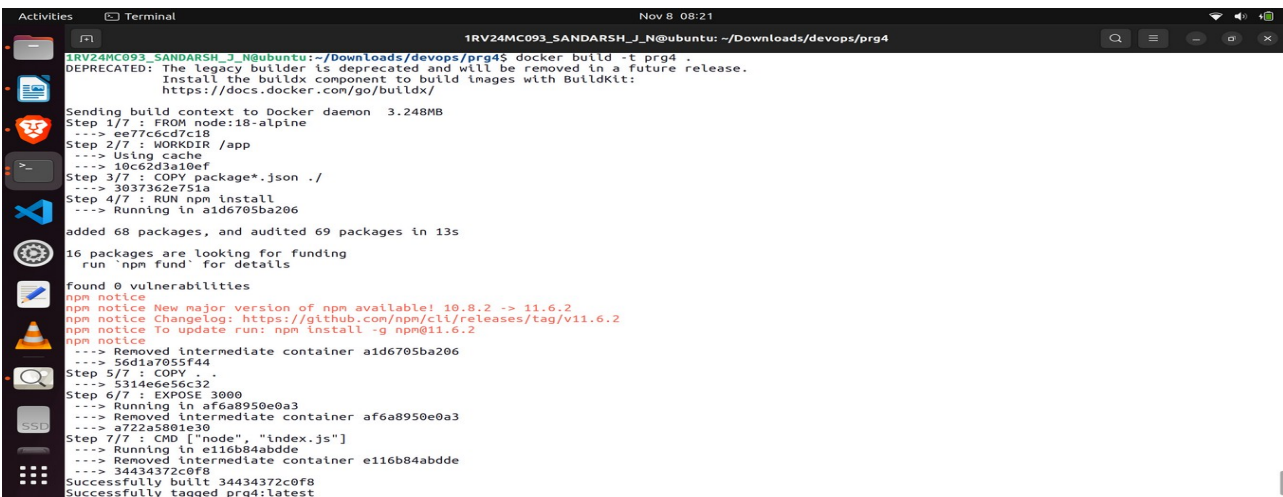
Srep 2 : create index.js and Dockerfile

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ nano index.js
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ cat index.js
const express = require('express');
const app = express();
const PORT = 3000;

app.get('/', (req, res) => {
  res.send('Hello from Git & Docker integrated Express application!');
});

app.listen(PORT, () => {
  console.log(`Express server running at http://localhost:${PORT}`);
});
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ nano Dockerfile
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ cat Dockerfile
FROM node:18-alpine
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3000
CMD ["node", "index.js"]
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$
```

Step 3 : create a docker image



```
Activities  Terminal  Nov 8 08:21
1RV24MC093_SANDARSH_J_N@ubuntu: ~/Downloads/devops/prg4

1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ docker build -t prg4 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.248MB
Step 1/7 : FROM node:18-alpine
----> ee77c6cd7c18
Step 2/7 : WORKDIR /app
----> Using cache
----> 10c62d3a10ef
Step 3/7 : COPY package*.json ./
----> 3037362e751a
Step 4/7 : RUN npm install
----> Running in aid6705ba206

added 68 packages, and audited 69 packages in 13s

16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.6.2
npm notice ChangeLog: https://github.com/npm/cli/releases/tag/v11.6.2
npm notice To update run: npm install -g npm@11.6.2
npm notice
----> Removed intermediate container aid6705ba206
----> 5ed1a7055f44
Step 5/7 : COPY . .
----> 5314ee56c32
Step 6/7 : EXPOSE 3000
----> Running in af6a8950e0a3
----> Removed intermediate container af6a8950e0a3
----> a722a5801e30
Step 7/7 : CMD ["node", "index.js"]
----> Running in e116b84abdde
----> Removed intermediate container e116b84abdde
----> 34434372c0f8
Successfully built 34434372c0f8
Successfully tagged prg4:latest
```

Step 4: initialize git

```
Activities Terminal Nov 8 08:32
1RV24MC093_SANDARSH_J_N@ubuntu: ~/Downloads/devops/prg4
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ ls
Dockerfile index.js node_modules package.json package-lock.json
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1RV24MC093_SANDARSH_J_N/Downloads/devops/prg4/.git/
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ git add .
```

Step 5 : commit the changes to git

```
Activities Terminal Nov 8 08:32
1RV24MC093_SANDARSH_J_N@ubuntu: ~/Downloads/devops/prg4
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ git commit -m "pushing prg4"
[master (root-commit) da24866] pushing prg4
636 files changed, 73039 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/README.md
```

Step 6 : push it to git

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ git remote add origin https://github.com/Sandarsh18/program4.git
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/Sandarsh18/program4.git'
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/Sandarsh18/program4.git'
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ git branch -M main
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$ git push -u origin main
Enumerating objects: 722, done.
Counting objects: 100% (722/722), done.
Delta compression using up to 16 threads
Compressing objects: 100% (667/667), done.
Writing objects: 100% (722/722), 733.90 KiB | 13.59 MiB/s, done.
Total 722 (delta 140), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (140/140), done.
To https://github.com/Sandarsh18/program4.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg4$
```

